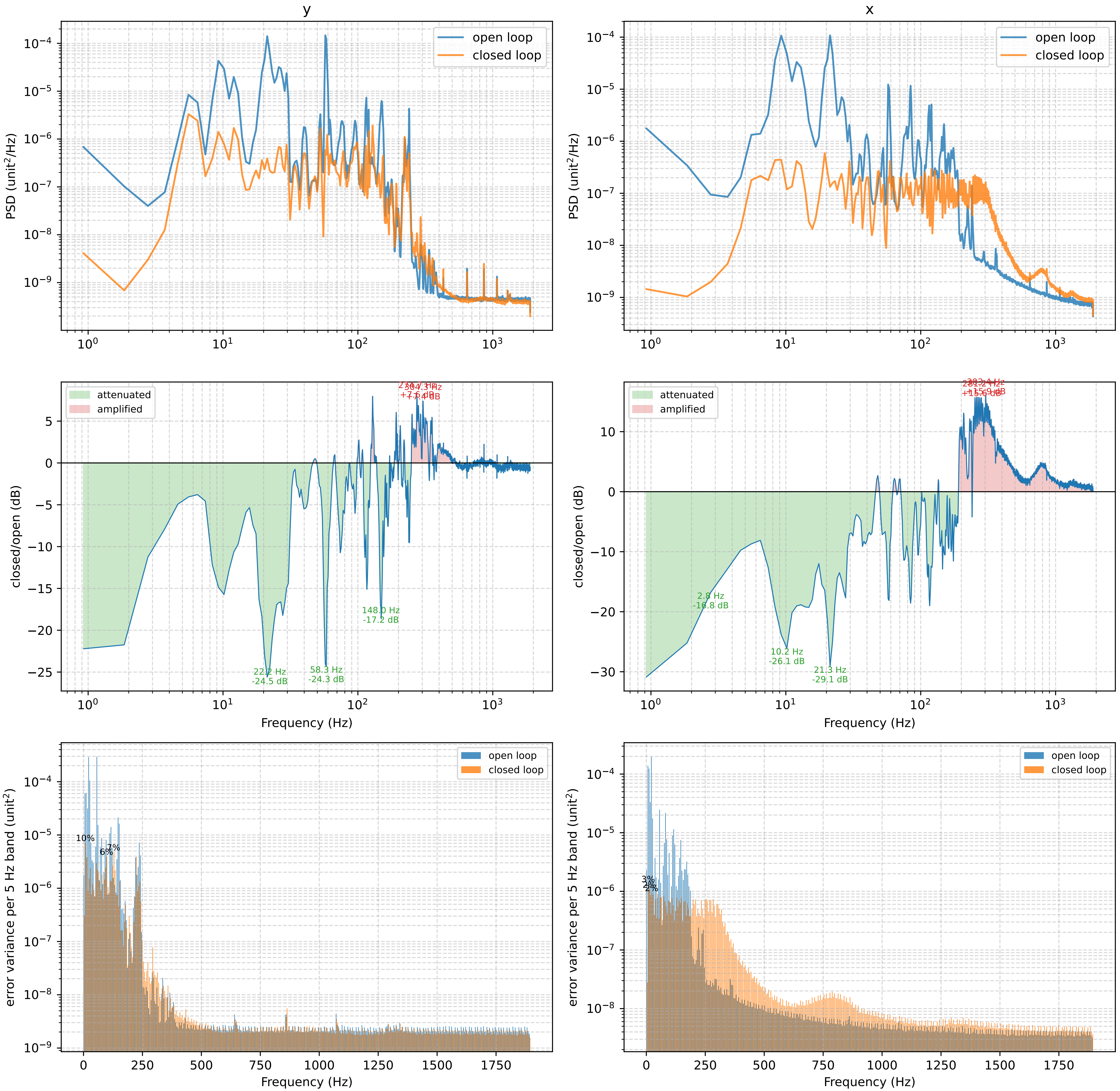
Closed-loop attenuation

y

x

Run configuration and per-band RMS

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Started: 2026-07-31 12:46:35
pixel_scale: 15.5 px per output unit
y: mean +5.948 -> -0.000 px;  RMS about mean 0.552 -> 0.129 px   (open -> closed)
x: mean +0.436 -> -0.000 px;  RMS about mean 0.423 -> 0.102 px   (open -> closed)

Config used:
ATTENUATION PAR-4 -- CONVERGED baseline at the CORRECTED small-signal Kx: PAR-3's controller with settle_time raised 120 -> 600 s and Kx re-fit 3.787 -> 4.05, through the PARPORT DAC (TI DAC81404 over the StarTech PEX1P2 / AX99100 PCIe parallel card, spi link, mmio backend), replacing the DAQC2/BRIDGEplate serial path used by every previous attenuation run.
TRANSPORT: bit-banged 24-bit SPI, ~6.4 us per channel update (~12.8 us for both channels), against the BRIDGEplate's ~0.5-1.4 ms. DAC range is FIXED at +-5 V bipolar (153 uV/LSB) with the mirror parked at 0 V, matching the bode sweeps exactly; auto-ranging is off, so the actuator is a fixed-gain element for both phases of the A/B and the open/closed comparison is clean. Supplies +/-12 V: AVDD needs VMAX+1.5 = 11.5 V for +-10, AVSS -11.5 V, leaving only 0.5 V of margin -- meter the rails under load. AVSS is therefore NEGATIVE, not grounded, which means the 0-5 base range runs outside the AVSS = 0 V condition SLASEH2A 7.5 gives the unipolar figures at (see _avss_tradeoff); 16-bit = 76 uV/LSB in 0-5, 305 uV/LSB in +-10, vs the BRIDGEplate's 12-bit 1 mV.
OUTPUTS: two channels, no coarse stage -- ch0 = y/tip (+1.0 V/unit), ch1 = x/tilt (-1.0 V/unit), offset 0 V, symmetric command bounds +-1 V (59 px of authority against a +-16 px ROI; the mirror's full range is deliberately NOT given to the loop on this first run). NOTE the command->FSM voltage mapping is NOT the same as any previous attenuation run, so K is not comparable across the transport change and was re-measured -- and it came out 4.7-5.5x HIGHER, not lower as this file originally predicted, so nothing about the piplate-era gains carries over.
Camera/CoM: 32x32 ROI start_x 1816 / start_y 684, exposure 50, gain 100, bw 100, 20x20 tracked window thr 5 (the camera is now shrouded, so the background floor no longer needs a high threshold). Identical to conf_bode_par_x/y[_small].json, so K and tau were fit through exactly this optical chain.
FSP: broad_order 512, broad_mu .03 -- REVERTED from PAR-2's 128/.005, which was 60% worse on y and 10% on x once normalised for the environment. 128 taps = 29.6 Hz resolution cannot separate the 2-3 Hz-spaced 9-30 Hz comb; this reproduces the documented kalman order-80 failure (order 80 ~29 Hz could not represent the 2.256 Hz comb). broad_lp 5 (retuned: the shorter delay moved the regeneration band from ~325 Hz to ~700 Hz); adapt_period 8, adapt_tau 5. Guard ON (guard_ratio 4). Cold-start observer: pass_open false + broad_freeze_closed false, so no weight is derived from open-loop data; drift_tau and the transient path are OFF. Modal freqs re-derived from run 1's own open PSD (8-mode cap), though they are inert while broad_order > 0.
PURPOSE: two defects in PAR-3, both fixed here. (1) The residual was still falling 500 s after closure (y asymptote 0.0404 px, tau 255 s), so its 500 s average scored a mostly-unconverged controller -- settle_time is now 600 s. (2) Kx was the +-100 mV figure 3.787 while the loop commands ~20 mV peak, where the plant reads 4.05, so x was over-commanded ~7%; that is the sign PAR-3 itself showed (x disturbance 28% above the open-phase error, x closed-window bound above the open-window one) and y, which is linear over 8-100 mV and did not move, is the control. This is the first number worth tuning against. Prior 500 s averages: PAR-1: y 0.518 -> 0.0804 px, x 0.496 -> 0.0966 px. PAR-2 (128/.005), normalised to PAR-1's environment: y 0.128, x 0.106; 60-150 Hz only 1.5x and the largest closed contributor on both axes; waterbed 3.4-3.5x at 195-250 Hz; no regeneration (400-800 Hz = 0.0% of command power).
DELAY AND K: taken from different sweeps on purpose. DELAY from the newest full 5-300 Hz sweeps, fsm_bodex_par2 / fsm_bodey_par2 (2026-07-31 12:23 / 12:16): x 2.763 fr (tau 0.4654 ms), y 2.624 fr (tau 0.4288 ms) at fs 3788. PAR-1..3 ran 2.717 / 2.577; the +0.046 fr shift sits inside the fits' own phase residuals, so the two sweep pairs agree and the newer/longer value is kept only because under-modeled delay is the documented failure mode. K from the SMALL-SIGNAL plateau: Kx 4.05, Ky 3.646. That distinction is not academic -- the loop commands ~6 mV rms / ~20 mV peak and the FSM compresses ~4% at +-100 mV, but WHICH AXIS compresses moves between days: on 07-30 it was x (100/50 mV ratio 1.041, y flat at 0.998), on 07-31 it was y (ratio 1.039, x flat at 1.003). The raw 07-31 fit lines therefore read Kx 4.050 (correct) and Ky 3.510 (3.7% low), and PAR-4 uses 3.646 for y from the +-20 mV sweep, corroborated by the same full file's +-50 mV pass at 3.662. Conversely tau is NOT measurable from a small sweep -- 10-30 Hz spans ~7 deg of phase and their loop_frames scatters 1.09-2.02 fr -- which is why delay comes only from the full sweeps. Against the piplate era this is ~5x MORE loop gain and LESS THAN HALF the delay; the shorter delay is the entire reason to expect better rejection here. Caveat: the small-signal gain itself wanders a few percent per hour (seven +-20 mV x sweeps read 3.97-4.36, each internally consistent), and erring HIGH on K is the safe direction since u = -phi_hat/K.
COMPARE against attenuation_fsp21 and attenuation_fsp15 (run 15: y 0.073 / x 0.168 px): report total and per-band closed RMS, 60-400 Hz waterbed, command/clamp behavior, and transient activation count on the shared 500/500/4096 grid. Closed RMS in PIXELS remains comparable across the change; command units do not. refs: Kulcsar 2006 / Petit 2008 / Meimon 2010 / Correia 2010
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RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.1212	0.0422	0.1837	0.0181
10-20	0.1496	0.0336	0.1944	0.0215
20-30	0.3070	0.0306	0.2273	0.0209
30-40	0.0503	0.0217	0.0351	0.0165
40-50	0.0296	0.0198	0.0244	0.0173
50-60	0.2685	0.0354	0.0795	0.0163
60-70	0.0635	0.0284	0.0199	0.0163
70-80	0.0594	0.0256	0.0475	0.0190
80-90	0.0277	0.0224	0.0842	0.0152
90-100	0.0567	0.0390	0.0391	0.0181
100-110	0.0192	0.0212	0.0211	0.0169
110-120	0.0773	0.0257	0.0699	0.0146
120-130	0.0418	0.0422	0.0424	0.0172
130-140	0.0218	0.0264	0.0263	0.0172
140-150	0.0772	0.0170	0.0518	0.0183
150-160	0.0653	0.0189	0.0302	0.0153
160-170	0.0140	0.0077	0.0337	0.0160
170-180	0.0147	0.0131	0.0350	0.0166
180-190	0.0083	0.0072	0.0232	0.0134
190-200	0.0078	0.0103	0.0077	0.0172
200-210	0.0050	0.0056	0.0055	0.0178
210-220	0.0164	0.0148	0.0063	0.0144
220-230	0.0364	0.0337	0.0091	0.0139
230-240	0.0483	0.0241	0.0076	0.0170
240-250	0.0321	0.0127	0.0080	0.0160
250-260	0.0026	0.0041	0.0037	0.0184
260-270	0.0024	0.0033	0.0035	0.0168
270-280	0.0020	0.0039	0.0036	0.0166
280-290	0.0017	0.0029	0.0035	0.0177
290-300	0.0034	0.0050	0.0039	0.0153
total 0-300	0.4927	0.1276	0.3966	0.0929